Jiaqin Wu

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EDUCATION

Georgetown University **Washington, DC**

Master of Science in Data Science for Public Policy *Aug. 2022 – May 2024*

Relevant Coursework: Statistics, Machine Learning, Deep Learning, Natural Language Processing, Time Series Analysis, Massive Data, Data Visualization

University of International Business and Economics **Beijing, China**

Bachelor of Management in Customs Administration *Sept. 2018 – Jun. 2022*

SKILLS

Programming: Python, R, SQL (MySQL, MS SQL Server, PostgreSQL), PySpark, JavaScript, STATA, SAS, HTML/CSS, Bash/Shell scripting, Excel

Machine Learning: Scikit-learn, XGBoost, LightGBM, CatBoost, Random Forest, Neural Networks, SVM, PCA, NLP, sentiment analysis, SHAP

Visualization & GIS: Tableau, Power BI, ArcGIS, Folium, GeoJSON, Plotly, Streamlit, Dash, R Shiny, Flask

Cloud & Data Tools: GCP, AWS (S3, Sagemaker, IAM, Containers), Azure, Docker, Git/GitHub, REST APIs, ETL pipelines

PROFESSIONAL EXPERIENCE

Center for Global Health Practice and Impact, Georgetown University | Python, R, MySQL, Streamlit, R Shiny, GCP **Washington, DC**

Data Scientist *Mar. 2023 – Present*

- Designed, validated and implemented **cross-departmental relational databases** in **MySQL** for 200+ medical institutions, managing 1K+ tables.
- Built and automated **ETL data pipelines** for 3M+ electronic medical records, improved data extraction, cleaning, quality assurance and reporting efficiency by **70%**.
- Processed, transformed, and analyzed 200K+ patient-level records using **Python** and **SQL** to support clinical cohort construction, descriptive epidemiology, risk stratification, and programmatic decision-making.
- Developed **reproducible data quality workflows** to identify missingness, inconsistent patient histories, implausible values, duplicate records, and longitudinal data gaps in national HIV electronic medical record datasets.
- Trained, validated and optimized **machine learning models** (Random Forest, XGBoost, CatBoost, LightGBM, Neural Networks, SVM) to predict HIV retention outcomes among people living with HIV, achieving **F2 score of 0.61** and **ROC-AUC of 0.90** using hyperparameter tuning and 5-fold cross-validation.
- Integrated the best-performing model (**CatBoost**) with **MySQL** databases and deployed an interactive **Streamlit** dashboard, providing healthcare professionals with patient-level risk predictions for targeted follow-up and differentiated care.
- Developed **REST APIs** for treatment interruption risk prediction, enabling scalable model deployment and integration with local healthcare systems.
- Used **Google Cloud Platform** tools to support secure data storage, cloud-based workflow execution, and scalable analytics infrastructure for global health data projects.
- Deployed an interactive **R Shiny** dashboard to visualize patient-level descriptive analyses, monitor HIV program indicators, and support real-time review of cohort trends.
- Built a **Python-based stakeholder portal** for technical assistance request submission, tracking, automated email notifications, backend updates, and coordination workflow management.
- Developed automated PDF report generation tools within **Streamlit** dashboards and implemented cloud-based file sharing, automated email workflows, and digital signature functionality to streamline technical assistance documentation.
- Contributed to peer-reviewed manuscripts, conference abstracts, and research presentations focused on machine learning, HIV treatment interruption, electronic medical record analytics, and global public health decision support.

Office of Community Safety, Prince William County | Python, Power BI, ArcGIS, SQL, JavaScript, Streamlit **Woodbridge, VA**

Data Analyst *Oct. 2024 - Present*

- Developed and deployed **Power BI dashboards** integrating data from Fire&EMS, Criminal Justice, Community Services and other county agencies to support inter-agency collaboration, service gap identification, and resource allocation.
- Built and managed an **MS SQL Server database** consolidating 100+ internal and external datasets and automated **Python ETL pipelines** for data cleaning, transformation, metric creation and dashboard refreshes, reducing manual processing time by **60%**.
- Applied **principal component analysis** correlation analysis, and domain-specific weighting methods in Python to construct the **Community Opportunity Index** across 92 census tracts, providing policymakers with a multidimensional measure of structural safety, service access, and community conditions.
- Integrated public datasets and county administrative data to assess socioeconomic stability, public health and service utilization, housing stability, mobility and transportation access, transportation safety, and environmental exposure at the census-tract level.
- Developed **interactive geographic visualization** of Community Opportunity Index using **ArcGIS, Streamlit** and **web mapping tools**, enabling policymakers and residents to identify underserved areas and prioritize targeted interventions.
- Analyzed countywide parking patterns using geospatial and administrative data in **Python** to identify intervention areas, geographic disparities, enforcement patterns, and policy-relevant recommendations.
- Built **Streamlit dashboards** to monitor **domestic violence trends** and **referral system activity**, providing cross-agency partners with real-time visibility into service usage and improving coordinated response.
- Created a **Community Resource Web Application** using **Python (Flask, Folium, Geopy, Javascript)**, enabling residents to search and locate food assistance, housing support, legal aid, healthcare and other essential county services through dynamic maps and geolocation tools.
- Supported county grant narratives, strategic planning, and cross-agency presentations by translating statistical and geospatial findings into actionable policy insights for leadership and community stakeholders.

Institute for Policy Studies (IPS) | Python, R, GitHub Actions **Washington, DC**

Consultant *Apr. 2026 – Present*

- Scraped and processed 120K+ Massachusetts housing transaction records using **Selenium** and **Python**, expanding a property tax analysis pipeline to support broader statewide fiscal impact modelling.
- Built a stakeholder-facing **Streamlit dashboard** with **Google Maps API** integration, allowing users to filter properties by geography, assessed value, transaction characteristics, and policy-relevant criteria.
- Automated daily extraction of the Forbes billionaire list using **Python** and **GitHub Actions**, scheduling recurring workflow and distributing refreshed outputs to stakeholders via automated email notifications.
- Standardized the web scraping workflow by structuring **Python scripts**, output datasets, logs, and documentation into a reusable pipeline, reducing manual refresh work and improving maintainability for future data updates.

Data Analyst *Jun. 2023 – Sept. 2023*

- Scraped 20K+ Boston housing transaction records using **Selenium**, cleaned and structured the data, and developed visualizations to estimate the fiscal impact of a proposed housing tax on properties valued above $2M.
- Created interactive county-level maps using **Google Maps API** and **GeoJSON** to display property distribution, assessed value patterns, and estimated tax exposure.
- Analyzed private flight data using FlightAware API data to estimate emissions impacts from the proposed Hanscom Airport expansion.
- Translated analysis results into policy-relevant findings and recommendations for local government, advocacy, and public policy stakeholders.

Baker Donelson | Python, Excel, Steamlit **Washington, DC**

Data Analyst Consultant *Sept.2024 – Oct. 2024*

- Automated **Python pipelines** to extract, clean and integrate loan datasets (Attom, HMDA, CDFI), reducing manual processing time by **50%**.
- Analyzed CDFI loans patterns, identifying states with the highest average loan amounts and quarterly peak lending periods, and built a **Streamlit dashboard** with dynamic filters and export features, enabling stakeholders to explore trends interactively and generate customized reports.

Center for International Private Enterprise (CIPE) | Python, AWS, NLP **Washington, DC**

MEL (Monitoring, Evaluation, Learning) Analyst *Sept.2023 – Dec. 2023*

- Built automated **ETL pipelines** on **AWS (S3**, **Lambda, Step Functions**) to extract, process and organize program data, reducing reporting delays.
- Applied **NLP (NLTK, Pandas)** on 2K+ abstracts and assess content themes, sentiment, and stakeholder engagement.
- Used **AWS Textract** and **Layoutparser** to digitize handwritten documents and convert unstructured records into structured CSV datasets for monitoring and evaluation analysis.
- Supported program evaluation by transforming qualitative and administrative data into structured evidence for learning, reporting, and stakeholder decision-making.

TEACHING EXPERIENCE

McCourt School for Public Policy, Georgetown University **Washington, DC**

Instructional Associate, Data Visualization *Aug. 2024 – Dec. 2024*

- Assisted in teaching a graduate-level data visualization course using Python, R, R Shiny and Tableau.
- Supported course delivery by designing course materials, grading assignments, holding office hours, and providing individualized student guidance.
- Advised students on best practices in visual analytics, data storytelling, dashboard design and communicating quantitative findings to audiences.

PUBLICATIONS

- **Wu, J**., Shaw, B., et al. *Applying machine learning to predict loss to follow-up among people living with HIV in Haiti using a national electronic medical record cohort.* International Journal of Public Health. https://doi.org/10.3389/ijph.2026.1609496
- Yu, W., Dawer, A., Floyd, J., Saad, N., **Wu, J.**, et al. *Opportunities and challenges in access to healthcare for international migrants with work-related diseases and injuries in Gulf Cooperation Council countries: A systematic literature review protocol.* PLOS ONE. https://doi.org/10.1371/journal.pone.0321681
- Kohli, A., Shaw, B., Greenberg, J., DeGraw, E., Habumugisha, L., Sebareze, J. L., Setubi, A. F., **Wu, J**., & Yantis, C. *Intimate partner violence and parents' use of violent discipline against young children in Rwanda: A baseline study of the REAL Fathers Initiative.* Children and Youth Services Review. https://doi.org/10.1016/j.childyouth.2025.108633

CONFERENCE PRESENTATIONS

- E-poster: "*An Innovative Community Engagement Model for Improving HIV Prevention: The LISTEN Process in Eswatini."* International AIDS Society Conference, 2025.
- E-poster: "*Predicting ART Interruptions: A Machine Learning Approach among PLHIV in Haiti*." International AIDS Society Conference, 2025.
- Poster: "*Applications and Interventions of Artificial Intelligence in Global Public Health: A Multinational Scoping Review*." Consortium of Universities for Global Health, 2025.
- Poster: "*Machine Learning for the Future of Global Health: A Case Study on Predicting HIV Treatment Interruptions in Haiti.*" Consortium of Universities for Global Health, 2026.
- Oral Presentation: "*Predicting Virological Failure in Pediatric and Adolescent Human immunodeficiency virus (HIV) Patients in Haiti: A Crisis-Adjusted Machine Learning Approach.*" Caribbean Public Health Agency 70th Annual Health Research Conference, 2026.

ACADEMIC & PROFESSIONAL SERVICE

- Abstract Reviewer, *Union World Conference on Lung Health, International Union Against Tuberculosis and Lung Disease*, 2026,
- Ad hoc Peer Reviewer, *Frontiers in Public Health*, 2026.